2. NON – UNIFORM BENDING – VERIFICATION OF RELATION BETWEEN LOAD AND DEPRESSION USING PIN AND MICROSCOPE

AIM

To verify the relation between the load and depression using non-uniform bending of a beam.

APPARATUS REQUIRED

A long uniform beam (usually a metre scale), two knife – edges, mass hanger, slotted masses, pin and vernier microscope.

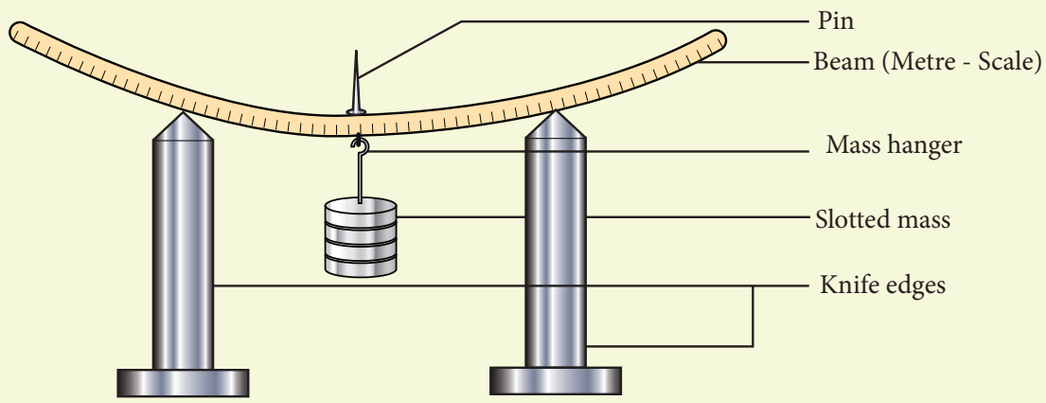
FORMULA

$\frac{M}{s}$ = a constant

where $M \leftarrow$ Load applied (mass) (kg)

s ← depression produced in the beam for the applied load(m)

DIAGRAM



EXPERIMENTAL SETUP OF NON - UNIFORM BENDING PIN AND MICROSCOPE

PROCEDURE

- Place the two knife – edges on the table.
- Place the uniform beam (metre scale) on top of the knife edges.
- Suspend the mass hanger at the centre. A pin is attached at the centre of the scale where the hanger is hung.
- Place a vernier microscope in front of this arrangement
- Adjust the microscope to get a clear view of the pin

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